MIL-A-10840D
17 November 1982
SUPERSEDING
MIL-A-10840C
30 June 1969

MILITARY SPECIFICATION

ARSENIC TRIOXIDE, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one technical grade of arsenic trioxide (As_2O_3).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-D-723	-	Drums, Fiber
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-60	-	Tape: Packaging, Waterproof

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-1168	-	Ammunition Lot Numbering

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179	-	Department of Transportation Hazardous Materials Regulations
29 CFR 1910.1018	-	OSHA Safety and Health Standards

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "49 CFR 171 to 179" and "29 CFR 1910.1018.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

C562	-	Moisture In Graphite
D1193	-	Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Appearance. Arsenic trioxide shall be a white powder when tested as specified in 4.2.4.1.

3.2 Assay. Arsenic trioxide shall have an assay of no less than 98.0 percent by weight as As_2O_3 when tested as specified in 4.2.4.2.

3.3 Moisture. Arsenic trioxide shall contain no more than 0.10 percent by weight of moisture when tested as specified in 4.2.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the arsenic trioxide produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test (see 6.5). Sampling for test shall be conducted in accordance with table I. A representative specimen of approximately 1/4 pound (lb) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 4.0 percent defective:

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TABLE I. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Fiberboard packing aids not as specified
(f) Palletization missing or inadequate
(g) Marking incorrect, missing, or illegible

4.2.3.2 For test (see 6.5). Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1/4 pound. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests (see 6.5). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine approximately 25 grams (g) of the specimen for form and color.

4.2.4.2 Assay. Weigh to the nearest 0.1 milligram (mg) approximately 0.2 g of the specimen into a 500-milliliter (ml) Erlenmeyer flask. Add 25 ml of 5-percent sodium hydroxide solution and agitate until the specimen is dissolved. Make slightly acid with 1 to 1 sulfuric acid. Add sodium bicarbonate in small portions until effervescence ceases and then add approximately 1 g in excess. Cool to room temperature and add 1 g of potassium iodide and 5 ml of 0.2-percent starch solution as indicator. Titrate with 0.1N iodine solution to the first appearance of a blue color. Calculate the percent by weight As_2O_3 as follows:

$$\text{Percent by weight } As_2O_3 = \frac{4.946\ AB}{W}$$

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where: A = Milliliters of iodine solution used,
B = Normality of iodine solution, and
W = Weight of specimen in grams.

4.2.4.3 Moisture. Determine the moisture content of approximately 5 g of the specimen, weighed to the nearest milligram, in accordance with ASTM C562.

5. PACKAGING

5.1 Unit packing, level C. Arsenic trioxide shall be unit packed level C in a 5-lb or 25-lb quantity as specified (see 6.2).

5.1.1 Five-pound quantity. A quantity of 5 lb (+1 or -0 ounce) of arsenic trioxide shall be packaged in a metal can in accordance with Department of Transportation (DOT) regulations. The can shall conform to type VI, with full friction plug or slip cover closure of PPP-C-96 and shall be formed from tin plate. Exterior nonmetallic coating shall not be required. The entire seam between the cover and body of the slip-cover can shall be sealed by a strip of minimum 1-inch wide tape conforming to PPP-T-60, type and class optional.

5.1.2 Twenty-five-pound quantity. A quantity of 25 (+1/4 or -0) lb of arsenic trioxide shall be packed in a fiber drum conforming to type I, grade A, class 2 of PPP-D-723 and DOT Specification 21C. The drum shall be closed as specified in PPP-D-723.

5.2 Packing, level C.

5.2.1 Five-pound cans. Four cans of arsenic trioxide, packaged as specified in 5.1.1, shall be packed in a fiberboard box conforming to style RSC, type CF, class domestic, variety SW, grade 175 of PPP-B-636. Half-slotted fiberboard separators shall be used to form a snug-fitting compartment for each can. Fiberboard pads shall be inserted against the inner faces of the box where needed to form a tight pack. Fiberboard separators and pads shall be formed from material conforming to type CF, class domestic, variety SW, grade 125 of PPP-F-320. The box shall be closed as specified for closure of class domestic boxes in PPP-B-636 using method I or II.

5.2.2 Twenty-five-pound fiber drum. The 25-lb quantity, unit packed in a fiber drum as specified in 5.1.2, shall require no further protection for shipment.

5.3 Unitization. Identical quantities per pallet of identical unit packs of arsenic trioxide shall be palletized in accordance with MIL-STD-147. Provision for stacking and safe supporting of superimposed loads shall be given prime consideration.

5.4 Marking. Unit and shipping containers and pallet loads shall be marked in accordance with DOT regulations, MIL-STD-129, and as follows:

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DANGER
CONTAINS INORGANIC ARSENIC
CANCER HAZARD
HARMFUL IF INHALED OR SWALLOWED
USE ONLY WITH ADEQUATE VENTILATION OR RESPIRATORY PROTECTION

Do not breath dust.
Keep container closed.
Wash thoroughly after handling.

POISON

Call a physician.

FIRST AID: If swallowed, induce vomiting by sticking finger down throat or by giving soapy or strong salty water to drink. Repeat until vomit is clear. Never give anything by mouth to an unconscious person. If inhaled, remove to fresh air. If not breathing, give artificial respiration, preferably mouth-to-mouth. If breathing is difficult, give oxygen.

Each shipping container shall bear a DOT "Poison" label securely affixed thereto.

6. NOTES

6.1 Intended use. Arsenic trioxide is intended for use in the manufacture of arsenic compounds.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Unit quantity required (see 5.1)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.5 Sampling and testing precautions. This specification covers inspection of chemical material which is potentially hazardous to personnel. Arsenic trioxide is a carcinogen, poison B. All applicable safety rules, regulations and procedures must be followed in the sampling and testing of this material.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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